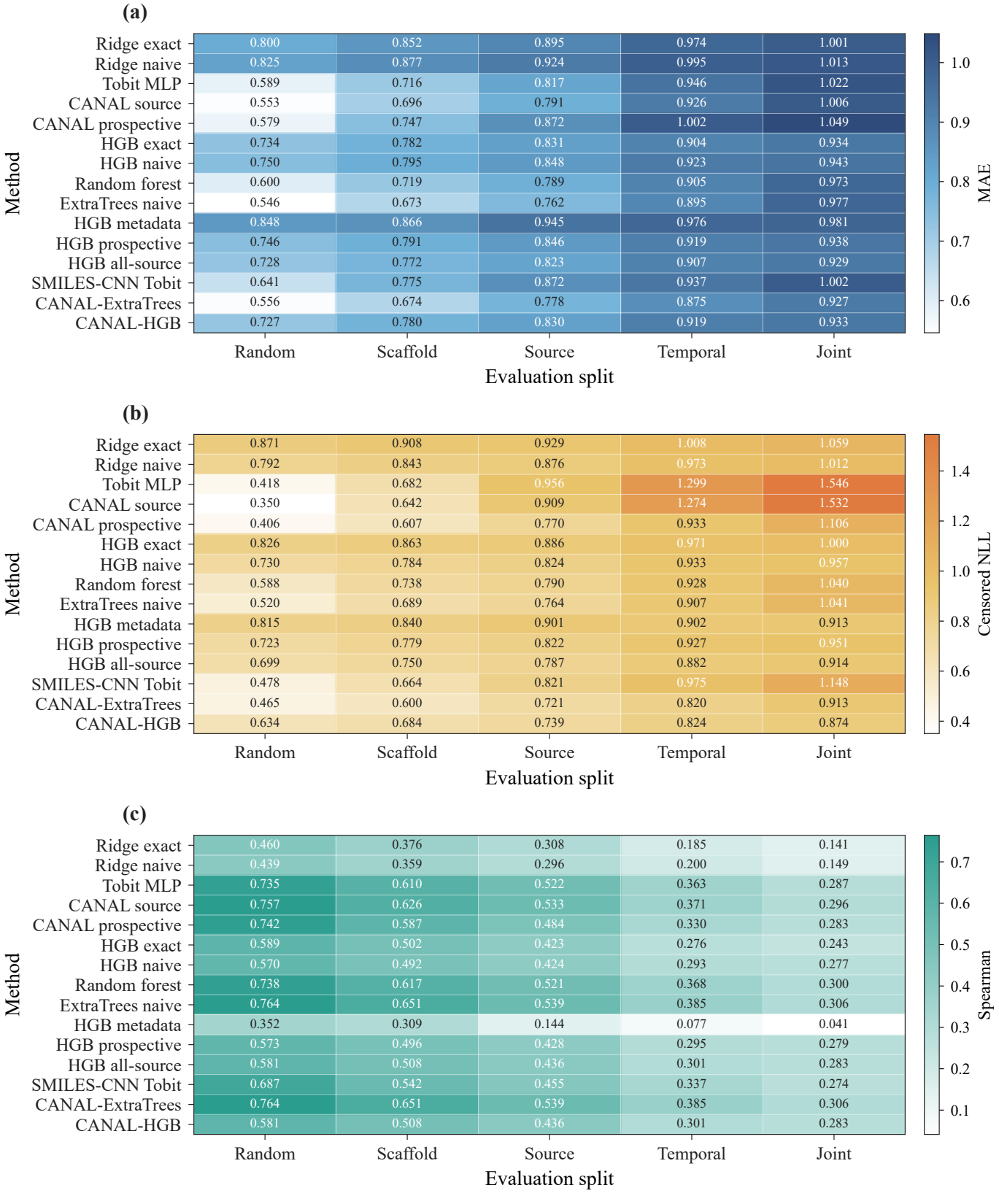

(a)
Ridge exact 0.800 0.852 0.895 0.974 1.001
Ridge naive 0.825 0.877 0.924 0.995 1.013
Tobit MLP 0.589 0.716 0.817 0.946 1.022
CANAL source 0.553 0.696 0.791 0.926 1.006
CANAL prospective 0.579 0.747 0.872 1.002 1.049
HGB exact 0.734 0.782 0.831 0.904 0.934
HGB naive 0.750 0.795 0.848 0.923 0.943
Random forest 0.600 0.719 0.789 0.905 0.973
ExtraTrees naive 0.546 0.673 0.762 0.895 0.977
HGB metadata 0.848 0.866 0.945 0.976 0.981
HGB prospective 0.746 0.791 0.846 0.919 0.938
HGB all-source 0.728 0.772 0.823 0.907 0.929
SMILES-CNN Tobit 0.641 0.775 0.872 0.937 1.002
CANAL-ExtraTrees 0.556 0.674 0.778 0.875 0.927
CANAL-HGB 0.727 0.780 0.830 0.919 0.933
Method
Random Scaffold Source Temporal Joint
Evaluation split
MAE 1.0 0.9 0.8 0.7 0.6
(b)
Ridge exact 0.871 0.908 0.929 1.008 1.059
Ridge naive 0.792 0.843 0.876 0.973 1.012
Tobit MLP 0.418 0.682 0.956 1.299 1.546
CANAL source 0.350 0.642 0.909 1.274 1.532
CANAL prospective 0.406 0.607 0.770 0.933 1.106
HGB exact 0.826 0.863 0.886 0.971 1.000
HGB naive 0.730 0.784 0.824 0.933 0.957
Random forest 0.588 0.738 0.790 0.928 1.040
ExtraTrees naive 0.520 0.689 0.764 0.907 1.041
HGB metadata 0.815 0.840 0.901 0.902 0.913
HGB prospective 0.723 0.779 0.822 0.927 0.951
HGB all-source 0.699 0.750 0.787 0.882 0.914
SMILES-CNN Tobit 0.478 0.664 0.821 0.975 1.148
CANAL-ExtraTrees 0.465 0.600 0.721 0.820 0.913
CANAL-HGB 0.634 0.684 0.739 0.824 0.874
Method
Random Scaffold Source Temporal Joint
Evaluation split
Censored NLL 1.4 1.2 1.0 0.8 0.6 0.4
(c)
Ridge exact 0.460 0.376 0.308 0.185 0.141
Ridge naive 0.439 0.359 0.296 0.200 0.149
Tobit MLP 0.735 0.610 0.522 0.363 0.287
CANAL source 0.757 0.626 0.533 0.371 0.296
CANAL prospective 0.742 0.587 0.484 0.330 0.283
HGB exact 0.589 0.502 0.423 0.276 0.243
HGB naive 0.570 0.492 0.424 0.293 0.277
Random forest 0.738 0.617 0.521 0.368 0.300
ExtraTrees naive 0.764 0.651 0.539 0.385 0.306
HGB metadata 0.352 0.309 0.144 0.077 0.041
HGB prospective 0.573 0.496 0.428 0.295 0.279
HGB all-source 0.581 0.508 0.436 0.301 0.283
SMILES-CNN Tobit 0.687 0.542 0.455 0.337 0.274
CANAL-ExtraTrees 0.764 0.651 0.539 0.385 0.306
CANAL-HGB 0.581 0.508 0.436 0.301 0.283
Method
Random Scaffold Source Temporal Joint
Evaluation split
Spearman 0.7 0.6 0.5 0.4 0.3 0.2 0.1